

A LOW-MEMORY FPGA-BASED SPECTRAL-CORRELATION ANALYZER
FOR DIGITAL QAM WAVEFORMS

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ABSTRACT

Cyclostationary signal processing (CSP) provides the ability to estimate received waveforms' statistical features blindly. Quadrature amplitude modulated (QAM) waveforms, when filtered by the square-root-raised cosine (SRRC) pulse shape function, have cyclic features that CSP can exploit to detect parameters such as Symbol Rate (SR) and Center Frequency (CF). The estimation of these SR-CF pairs allows for cognitive radio (CR) spectrum sensing techniques such as spectral awareness and interference mitigation. Here, we investigate a field-programmable gate array (FPGA) application of a blind symbol rate-center frequency estimator. First, this study provides a background on the theory behind cyclic spectral density function (CSD), spectral correlation analyzers (SCA), and spectrum sensing. Following this, an existing SCA is investigated, with commentary on how its algorithm can be improved is provided. Lastly, an FPGA SCA implementation for low-resource devices, such as CubeSats, is discussed. The paper concludes by reporting its performance characteristics.